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Solvent optimization for caffeine and tannic acid extraction from Guarana seed: Insights from molecular dynamics simulations

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ABSTRACT

Guarana seeds are a rich source of bioactive compounds, particularly caffeine and tannic acid, which have significant applications in the food and pharmaceutical industries. To extract these valuable compounds from the guarana seeds, solvent extraction methods are used. Molecular dynamics (MD) simulations are employed to investigate the interaction of Guarana seed extract with three different solvents namely ethanol, ethyl acetate and Choline chloride-glycerol as deep eutectic solvent (DES) varying the molar ratio. The bonded and non-bonded interactions are computed to perform the MD simulations. The optimized potentials for liquid simulations force field are used to compute these interactions. The study examines the Radial Distribution Function (RDF) and hydrogen bonds to elucidate structural arrangement and intermolecular interactions respectively. RDF reveals that the interaction of caffeine with ethanol is strongest hence it is the best suitable solvent for its extraction. Mean Square Displacement analysis offers insights into transport properties, particularly diffusion of molecules in solvents. Primarily subdiffusive nature for the mobility of functional moieties of Guarana seed components are observed in solvents. The comparison of diffusive regimes further reveals the preferential extraction of caffeine and tannic acid over the minor compounds using ethanol. The selective separation leads their essential applications in the food industry.

1. Introduction

Caffeine, a naturally occurring chemical stimulant found in various plant species, particularly in the leaves, seeds, and fruits, is a member of the trimethyl xanthine group with the chemical formula $C_8H_{10}N_4O_2$. Renowned for its role in enhancing mental alertness, caffeine finds diverse applications beyond its stimulant properties. One notable source of caffeine is Guarana (scientifically known as *Paullinia cupana* Kunth, Sapindaceae) [1], a plant native to the Amazon region, traditionally utilized by indigenous communities for its invigorating effects. With caffeine concentrations reaching up to 6 %, Guarana seeds have become a valuable reservoir of this alkaloid [1,2]. The presence of puranic alkaloid caffeine (1,3,7-trimethylxanthine), makes the Guarana seed particularly useful as a substance for stimulant source [4,5]. Other puranic alkaloids such as theophylline, theobromine [3,4] are also present in Guarana seed. Hence, the major value-added components of Guarana seeds are tabulated in Table S1 of Supplementary material. The extraction of caffeine from Guarana has spurred significant interest in the pharmaceutical and food industries [5] leading to the exploration of various extraction methods. The extraction methods and process parameters are found in literature [6,7] which primarily refer to ethanol mediated as well as non-alcoholic polysaccharide-based extract. Among these methods, solvent extraction emerges as a pragmatic and economically viable choice. Traditional solvents such as ethanol and ethyl acetate have been historically employed for caffeine extraction, but with advancements in extraction techniques, diverse methods such as accelerated solvent extraction, microwave-assisted extraction, supercritical fluid extraction, and ultrasound-assisted extraction have gained prominence [8–10].

Methylene chloride, ethyl acetate, ethanol along with water are commonly used as solvent for caffeine extract from Guarana seed [11]. A proportionate mixture of hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD) is commonly referred to as deep eutectic solvent (DES) has been viewed as state-of-the-art extractant of caffeine. Recently, caffeine extraction from coffee has been facilitated by choline chloride-sorbitol, betaine-sorbitol, choline chloride-glycerol [12,13]. One key challenge in extraction by experimental method is the presence of various compounds such as tannin, catechin, epicatechin alongside

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caffeine, requiring careful optimization of extraction methods to ensure purity. However, accurate quantification of extracted caffeine will be ensured despite potential interference. Solvent selection is a critical aspect of the extraction process, and the chosen solvent must possess specific characteristics for an efficient extraction. Factors such as immiscibility with water, low boiling point, non-toxicity, non-reactivity, and cost-effectiveness play crucial roles in determining the efficacy of a solvent. However, experimental methodology cannot alone reveal all such potential molecular level interactions that play a crucial role in extraction. Hence, complementing experimental methods, molecular dynamics (MD) simulations will help understand molecular interactions, estimate the structural and transport properties, identify key extraction factors, and thus accelerate the development value added component extraction from Guarana seeds.

Recognizing the challenges in experimental solvent selection, MD simulation is emerging as a valuable tool which gives better understanding of molecular level extraction process at the molecular level. MD simulations provide valuable cognizance into solute–solvent interactions, offering a microscopic understanding of the extraction and separation mechanisms [14]. While theoretical studies through MD simulations complement experimental approaches, they do not replace them. Instead, they enhance the understanding of complex conditions and facilitate exploration of new solvent systems. For instance, computer simulations are used to understand how ethanol extracts long-chain alcohols [15]. The separation of solvents into different layers from a mixture of biological buffer with organic solvents and water are also observed through MD simulation [16]. Notably, employing MD for extraction presents distinct advantages over traditional experimental methods, including cost-effectiveness and the ability to capture the atomistic dynamic phenomena. Through this approach, the research endeavours to pioneer novel insights into caffeine extraction processes, contributing to the advancement of efficient and sustainable extraction techniques.

The primary aim of this work is to select a suitable solvent for the extraction of caffeine and tannic acids from the guarana seeds. The molecular level non-bonded interactions during the extraction process, unravelling the dynamic chemistry at play [17–20]. The experimental literature shows that the extraction of caffeine and tannic acids from the guarana seeds using ethanol, ethyl acetate, and DES (choline chloride-glycerol) has been widely studied [21,22]. Moreover, these three solvents have different dipole moment values: ethanol (1.66 D), ethyl acetate (1.88 D) and DES (5.92 D) [23], which directly influence the polarity of the solvents. However, the interaction of various components of Guarana seed with solvents facilitating the extraction is still elusive [11,24,25]. To have clarity, MD study involving components of Guarana seed and three widely used solvents namely, ethanol, ethyl acetate, and choline chloride-glycerol as DES have been undertaken. Initially, the focus lies on identifying solvents best suited for this purpose, considering factors such as solubility and selectivity. Concurrently, the development of tailored force field models for different Guarana seed species aims to enhance the accuracy of MD simulations. These simulations delve into the structural and transport behaviours of various species within the extraction medium, offering detailed insights into molecular interactions and diffusion dynamics. A sequence of analysis with radial distribution function, hydrogen bonding and diffusivity promises to deepen the comprehension of governing the caffeine extraction process from Guarana seeds.

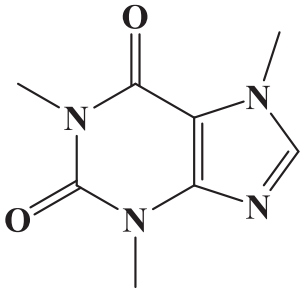
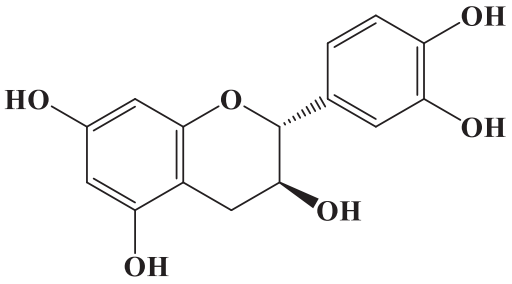
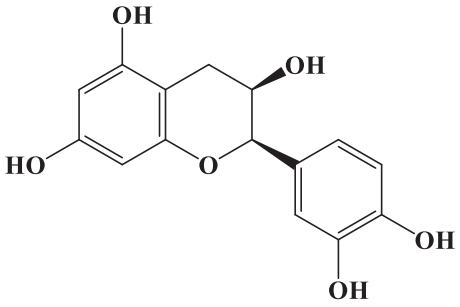
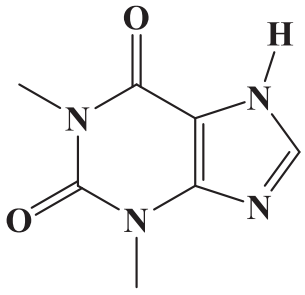
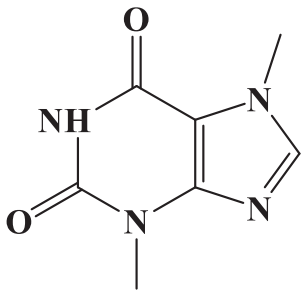
2. Computation methodology

Primarily, Guarana seed extract contains caffeine, theophylline, procyanidins, catechin, theobromine, epicatechin, and tannic acid [1]. The representative molecular structures along with the compositions in the seed extract are given in Table 1. Polar solvents such as ethanol, ethyl acetate, and neoteric solvents like DES consisting of glycerol and choline chloride, are selected for the in-silico extraction study. The structures are initially drawn in the Avogadro open-source package [26]. The Avogadro open-source package is further used to optimize the initial geometry. The initial geometry is optimized using the universal force field, which is presented in the Avogadro source code. After the geometry optimisation the PACKMOL package [27] is used to create the initial simulation boxes for MD simulation. The solute to solvent molar ratio has been varied as 1:9, 1:1 and 9:1. Such variation in molar ratio would reveal the solvent effect on solute extraction. The exact compositions of various systems coupled with initial simulation box dimensions are given in Tables S1–S2 of Supplementary materials. Optimized Potentials for Liquid Simulations (OPLS) force field has been proven effective force field for organic molecule extraction using polar and neoteric solvents [28] and henceforth is used in this work. The functional form of OPLS force field contains bonded (bond, angle, dihedral and improper) and nonbonded (6–12 LJ potential and coulombic) energies whose functional form is described in literature [29].

2.1. Methodology for MD simulation

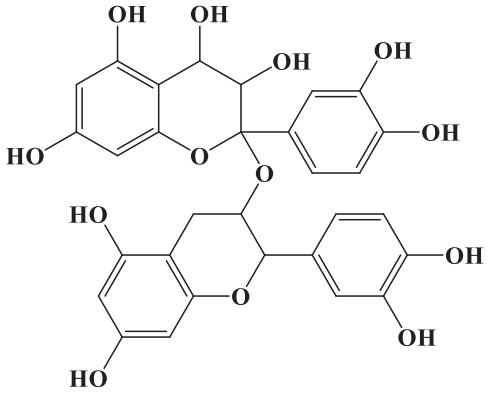
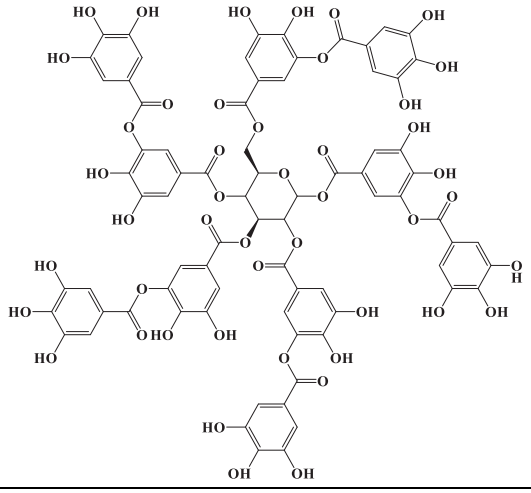
The MD simulation comprises energy minimization, equilibration (including both NVT and NPT ensembles), and a concluding production run. Every MD simulation utilizes the GROMACS package [30], and trajectory analysis employs the Visual Molecular Dynamics (VMD) package [31]. Energy minimization lowers the potential energy of the system, aiding molecules in attaining a stable conformation. Reducing the potential energy enhances the stability of the system. Energy minimization uses the Steepest Descent algorithm with a convergence threshold of 1000 kJ/mol nm. It is presently underway for 500,000 steps, with a step size of 0.001 nm. A leap-frog integrator [32] integrates all equations of motion using a time step of 2 fs during the equilibration and final production run. The choice of a 2-fs time step in all-atom MD simulations is based on its ability to accurately capture system dynamics while maintaining computational efficiency. The fastest motions typically involve bonds with hydrogen atoms, which oscillate rapidly. By applying constraints to these bonds, a 2-fs time step can be safely used without risking instability in the simulation. This approach balances accuracy and speed, effectively doubling simulation speed compared to a 1-fs time step widely accepted in simulation literature [29,33–35]. The 2-fs time step also aligns with Nyquist's theorem, ensuring proper capture of all necessary molecular motions [36]. The linear constraint solver (LINCS) algorithm [37] constrains hydrogen-containing moieties motion in the equilibration stage. The LINCS algorithm is specifically designed to handle large molecular systems with many degrees of freedom, making it particularly suitable for the study. It can maintain constraints on bond lengths and angles more efficiently, especially in complex simulations, without compromising accuracy. This efficiency is critical when performing extensive simulations, as it reduces computational time while ensuring reliable results. LINCS algorithm is widely used in literature for complex and similar types of systems [29,34,35,38]. The motion of hydrogen-containing moieties is restricted using the LINCS algorithm to stabilize the system while allowing other parts of the molecule to adapt to the environment. Converting bonds involving hydrogen atoms to constraints is a typical strategy in molecular simulations. The hydrogen atoms are lighter and more susceptible to rapid vibrations, which can lead to numerical instability. By constraining these bonds, the integrity of the molecular structure is maintained without significantly affecting overall dynamics, allowing for a larger time step and a more efficient equilibration process. The particle-mesh Ewald (PME) method [39] handles the long-range electrostatic interactions. van der Waals (vdW) interactions are limited to a distance of 1.2 nm. The Verlet algorithm updates the neighbor grid list at a frequency of 20 steps. Equilibration occurs in two stages: the first stage involves a canonical ensemble (NVT), and the second stage involves an isothermal isobaric ensemble (NPT). Throughout both the equilibration and final production stages, temperature remains constant via the

Table 1
System composition.

Compound name	Abbreviations	Composition (in mole %)	Structure	References
Caffeine	CAF	6		[22]
Catechin	CA0	0.7		[46]
Epicatechin	E0	1		[46]
Theophylline	T0	0.5		[47]
Theobromine	THO	1		[48]

(*continued on next page*)

Table 1 (*continued*)

Compound name	Abbreviations	Composition (in mole %)	Structure	References
Procyanidins	P0	2		[1]
Tanic Acid	TAN	15		[49]

velocity-rescaling thermostat (V-rescale) [40]. Additionally, the pressure remains controlled during the NPT and final production stages by employing the Parrinello-Rahman barostat [41]. A unit ns simulation with NVT ensemble is conducted at a temperature of 300 K with a time step of 2 fs. The objective of the NVT ensemble is to maintain the system's temperature at 300 K, which is shown in Figs. S1 and S2 of Supplementary materials. The simulation with NPT ensemble runs for 19 ns at a temperature of 300 K and a pressure of 1.013 bar, with a time step of 2 fs. The NPT equilibration outcome indicates that the energy, density, and volume of the system remain nearly constant. This finding suggests that the system is adequately equilibrated and ready for the production run. The final production run spans 30 ns at temperatures of 300 K and pressures of 1.013 bar. After analyzing the trajectory, the additional 100 ns production run is conducted in the best system under the same conditions to better understand the interactions between moieties. The Coordinates and velocities of the system are recorded at every ten ps for future analysis.

2.2. Analysis of trajectories

2.2.1. Hydrogen bonding (H bonding)

The presence of various electronegative atoms in the solute and polar solvents generate ample amount of hydrogen bonding (H-bonding) interactions which is presumed to play a dominant factor in extraction behaviour. The strength of H-bonds among caffeine and other bioactive compounds and solvents depends on various factors, including the type of solvent, the extent of interaction, and the presence of functional moieties capable of participating in hydrogen bonding. The formation of H-bond between an acceptor atom (A) and a donor atom (D) is determined using a geometrical criterion. Specifically, when the hydrogen donor-to-acceptor angle (α_{HDA}) is less than 30°, and the distance between the donor and acceptor atoms, i.e. r_{DA}, is less than 0.35 nm, a hydrogen bond is considered to have formed. To analyse the H-bonding, the intermittent hydrogen bond autocorrelation function is calculated using a defined criterion, providing insights into the temporal evolution of hydrogen bonds within the system.

$$C_{HB} = \frac{\langle H(\tau)H(\tau+t)\rangle}{\langle H(\tau)H(\tau)\rangle} \tag{1}$$

In this context, $H(t)$ is considered to be either 1 or 0, C_{HB} represents the count of hydrogen bonds, and depending on whether the selected pair forms a hydrogen bond at that particular moment in time.

2.2.2. Radial distribution function (RDF)

The radial distribution function (RDF) helps to understand the distribution of atoms surrounding a selected atom. Therefore, the structural information along with the atom–atom interactions can be inferred. The mathematical expression for the RDF is articulated as follows:

$$g(r) = \frac{1}{N}\left(\frac{V}{4\pi r^2 dr}\right)\sum_{i=1}^{N}\sum_{i\neq j}^{N}\delta\left(|r_i - r_j| - r\right) \tag{2}$$

Here, r is the distance between the reference particle and another particle in the system, V represents the total volume of the simulation box, $\delta(z)$ is the Dirac delta function (equal to 1 if $z = 0$ and 0 otherwise), dr is the width of the radial bin, and $(r_i - r_j)$ is the distance between particles i and j. The RDF computation entails dividing the simulation box into a series of concentric spherical shells with widths of dr. Within each shell, the particles are tallied, and the outcome is normalized by the

system's density to yield $g(r)$. The coordination number (CN) is further calculated from Eq. (3) using the RDF and it determines the number of particles present in various solvation shells.

$$CN(r) = 4\pi\rho N \int_0^r g(r) r^2 dr \tag{3}$$

The coordination number $CN(r)$ is determined by integrating the radial distribution function g (r) over a spherical shell of radius r. In this integral, N is the total number of particles in the system, ρ represents the particle density, and $g(r)$ is the radial distribution function. The result of this integration over the spherical shell, cantered at the reference particle, provides a count of particles within a specified distance from the reference particle. When adjusted correctly, this tally gives the coordination number, which is an important gauge for understanding the local environment and interactions within the system.

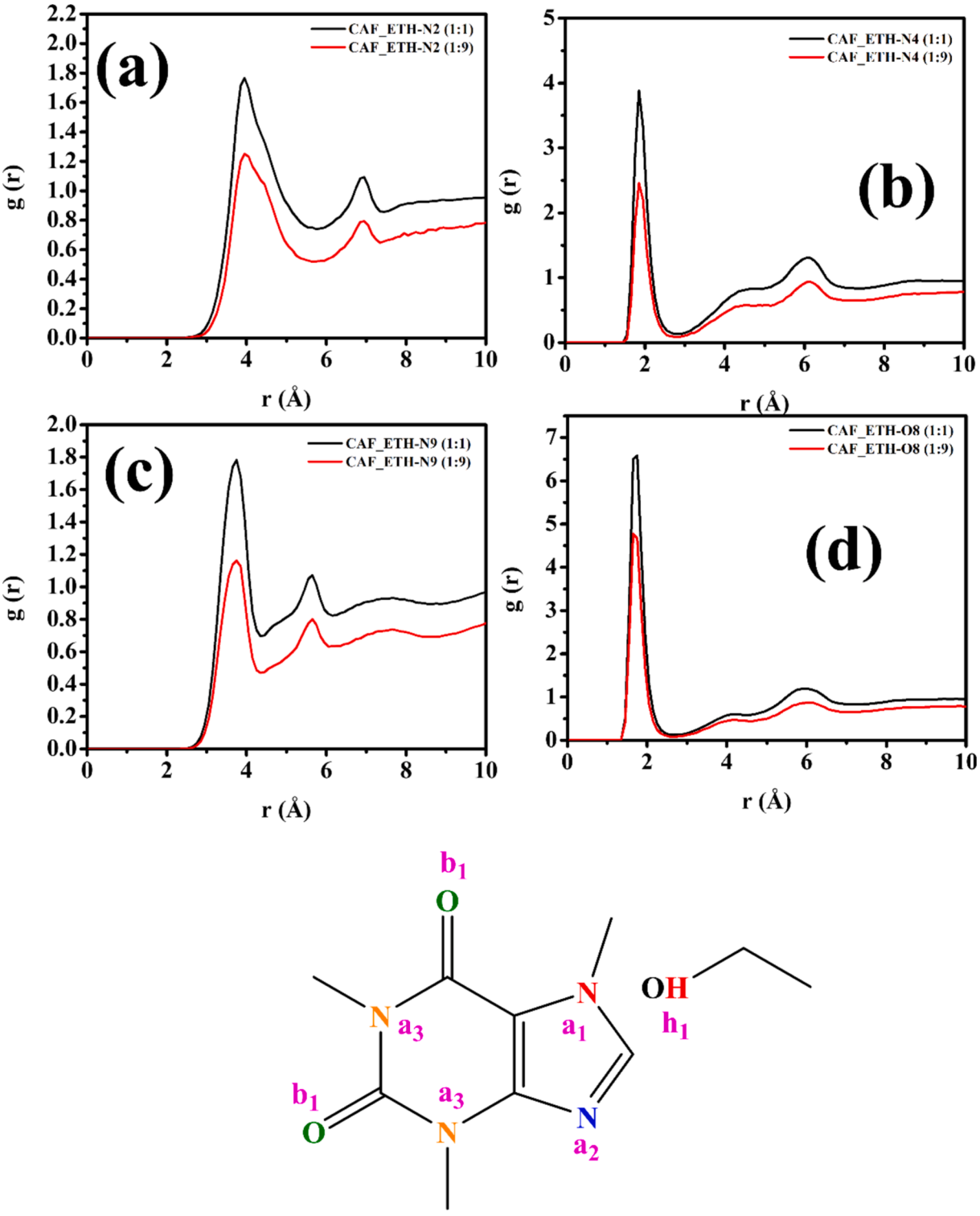


Fig. 1. RDF of caffeine in ethanol-based system; (a) N2—H (denoted by $a_1 \ldots h_1$), (b) N4—H (denoted by $a_2 \ldots h_1$), (c) N9—H (denoted by $a_3 \ldots h_1$) and (d) O8—H (denoted by $b_1 \ldots h_1$).

2.2.3. Mean square deviation (MSD)

The self-diffusion, a pivotal characteristic in extraction, is effectively assessed by plotting the slope derived from the mean squared displacement (MSD) of molecules over a sufficiently extended time span. The MSD characterizes the particle or molecule displacement from its initial position. Computation of MSD involves squaring the difference between the particle's position at a given time and its initial position, followed by averaging this value across all particles within the system. The subsequent equation is commonly employed for determining the MSD:

$$MSD(t) = \langle (r_i(t) - r_i(0))^2 \rangle = \frac{1}{N} \sum_{i=1}^{N} [(r_i(t) - r_i(0))^2] \quad (4)$$

In this equation, $r_i(t)$ signifies the position of atom i at time t, $r_i(0)$ denotes its initial position, and the angled brackets $\langle\ \rangle$ represent an ensemble average over all particles in the system. Leveraging the Einstein relationship [42,43] the gradient of the MSD can be employed for the computation of the diffusion coefficient (D).

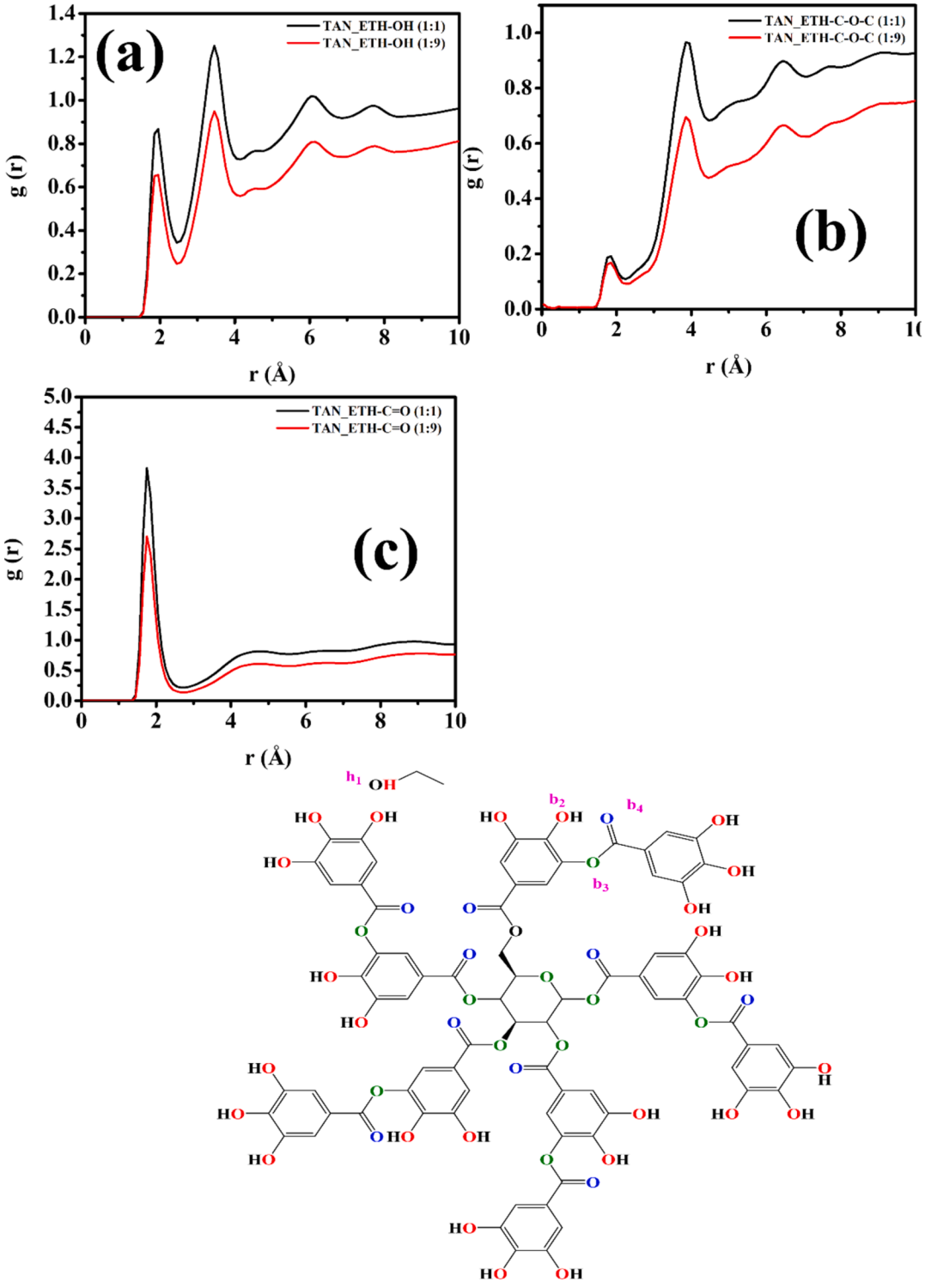


Fig. 2. RDF Tannic acid of ethanol-based system; (a) O—H (denoted by $b_2 \ldots h_1$), (b) C—O—C—H (denoted by $b_3 \ldots h_1$) and (c) C═O—H (denoted by $b_4 \ldots h_1$).

$$\beta = \frac{1}{6}\lim_{t\to\infty}\frac{d[\log MSD(t)]}{d[log(t)]} \quad (5)$$

$$D = \frac{1}{6N}\lim_{t\to\infty}\frac{d}{dt}\sum_{i=1}^{N}[(r_i(t) - r_i(0))^2] \quad (6)$$

3. Results & discussion

3.1. Radial distribution function (RDF)

3.1.1. Distribution of various components of Guarana seed in ethanol

Fig. 1 represents the RDF among various moieties of caffeine and ethanol being mixed in 1:1 (system I) and 1:9 (system II) molar ratios. The functional moieties involved in the RDF are denoted in the subplots too. Fig. 1a–c denotes the distribution of various nitrogen moieties with respect to the hydrogen of the —OH moiety of ethanol. A general observation from the three subplots refers to the increase in nonbonded interaction among the said moieties with the enhanced molar ratio of caffeine. The higher amount of caffeine (i.e. 1:1 mixture) creates a bigger number of available nitrogen and thus produces a bigger distribution envelope. The sharp nature of the RDF peak in the first solvation shell indicates a stronger vdW interaction. Hence, the first solvation shell from CAF_ETH_N2 resides in 5.85 Å whereas the same for CAF_ETH_N4 (denoted by a_2…h_1 in Fig. 1b) and CAF_ETH_N9 (denoted by a_3…h_1 in Fig. 1c) are 2.95 and 4.45 Å respectively. Among the four nitrogen atoms in caffeine, N2 (b denoted by a_1…h_1 in Fig. 1a) influences larger area and thus has CN of 4.8 and 3.9 for 1:1 and 1:9 M mixtures, respectively. However, the sharpness of RDF denotes the intensity of interaction among moieties. In that sense, the presence of immediate —CH_3 with N2 and N9 leaves the electron cloud of them less exposed. However, the electron cloud of N4 is relatively free to have non-bonded interaction with hydrogen of —OH of ethanol. However, for all nitrogen-related RDFs, the second solvation shell is prominent primarily in long-range interactions. Additionally, the oxygen atoms of caffeine can interact with the hydrogen of —OH moieties more intensely than their nitrogen counterpart. The first solvation shell is detected at 2.75 Å with *CN* of 0.7 and 0.6 for 1:1 and 1:9 M ratios, respectively. The peak values of RDF conclude that oxygen (denoted by b1 in Fig. 1d) and nitrogen (denoted by a_2 in Fig. 1b) primarily establish nonbonded interaction with ethanol.

Tannic acid is another important extract from Guarana seeds, which requires co-extraction using solvents such as ethanol. There are three oxygen-containing moieties in the tannic acid having ether, hydroxyl and carbonyl characteristics (denoted by b3, b2 and b4 in Fig. 2, respectively). The RDFs of these with the hydrogen of —OH moiety of ethanol is given in Fig. 2a–c. The speculation of dissociation of hydrogen from the —OH moiety is reflected by the RDF. The first peak of RDF of Fig. 2a is lower than the second peak relating to the weak interaction between TAN_ETH_OH (denoted by b_2…h_1 in Fig. 2a). Hence, the *CN*s of the first solvation shell are 0.15 and 0.19 for 1:1 and 1:9 M ratios, respectively, whereas the *CN*s for the second solvation shell are 1.2 and 1.62 for the same molar mixtures. The domination of the second solvation shell is further prominent for the TAN_ETH_C-O-C (denoted by b_3…h_1 in Fig. 2b). Such dominances indicate the shielding of electron clouds, which hinders H-bonding interactions. The prominent H-bonding interaction is observed by —CO moiety of tannic acid with —OH moiety of ethanol (TAN_ETH_C = O) (denoted by b_4…h_1 in Fig. 2c) and the corresponding RDF is given in Fig. 2c. The first solvation shell spreads up to 2.85 Å with a CN of 0.52 and 0.40 for 1:1 and 1:9 M ratios, respectively.

The other two prominent tannins, i.e. catechin and epicatechin, are present in small quantities on Guarana seeds. The various oxygen atoms in the epicatechin and catechin are able to form H-bond with the hydrogen of —OH moiety of ethanol. Figs. S3a and S4a depict the possible H-bond formation involving the ether moiety (denoted by d1 and e1, respectively) of catechin and epicatechin. However, the lower intensity of RDF in the first solvation shell diminishes such possibility; instead, there remains a strong possibility of intramolecular H-bonding in catechin and epicatechin as observed from Figs. S3b and S4b. However, stronger H-bond formation is observed between —OH moieties of catechin and epicatechin with ethanol, as observed in Figs. S3c–d and S4c–d. The solvation shell structure of trace compounds does not always follow the trail of major compounds, i.e., the envelope of RDF increases with the molar ratio of solvent. This further corroborates the greater affinity of caffeine and tannic acid moieties towards ethanol. Hence, selective extraction of them increases with ethanol, and the implication from the simulation resonates with the experimental observation.

Procyanidin, theobromine, and theophylline are found in very small quantities in Guarana seed. Despite not having a significant extraction possibility of these from Guarana seeds, the RDFs are summarized as they can have a meaningful influence in the extraction process. The oxygen of —OH moiety forms an H-bond with ethanol having the first solvation shell spreads up to 2.55 Å (Fig. S5a of Supplementary material). However, such non-bonded interactions are not visible for the other oxygen atoms present on the procyanidin (Fig. S5b–c of Supplementary material). The replacement of methyl moiety in the N2 position of caffeine generates the structure of theophylline. However, this replacement marks significant changes in the RDF, as can be seen in Figs. S5 and S6 of Supplementary material. The nitrogen atoms of the pentagon of theophylline do not produce significant interactions with the —OH moiety of ethanol. The nitrogens of the hexagon of theophylline produce H-bond interaction with —OH moiety of ethanol ending first solvation shell at 4.15 Å and CN of 1.16 and 0.99 for 1:1 and 1:9 M mixtures, respectively. Like the caffeine structure, a major H-bond interaction between theophylline and ethanol has been observed in Fig. S6d. This is in line with the behaviour of caffeine in ethanol. However, nitrogen atoms of theobromine produce very good H-bond interaction with the —OH moiety of ethanol, as seen in Fig. S7 of the Supplementary material. The presence of —CH_3 adjacent to nitrogen downgrades the availability of electron cloud for making effective H-bonding, as revealed by Fig. S7d of Supplementary material. However, relatively free electron clouds of nitrogen produce almost equal intensity and spread of RDF compares to the oxygen moiety (Fig. S7a–c).

3.1.2. Distribution of various components of Guarana seed in ethyl acetate

The absence of hydrogen atoms associated with the electronegative atoms of ethyl acetate leaves possible H-bond formation between ether and carbonyl moieties of ethyl acetate to the —OH moiety of tannic acid. For the same reason, ethyl acetate cannot make effective H-bonding with the functional moieties of caffeine, leading to poor extraction. Fig. 3 represents the RDF among hydrogen of —OH moieties of tannic acid and two different oxygen moieties (ether (O3) and carbonyl (O5)) (denoted by o_1 and o_2 respectively) of ethyl acetate being mixed in 1:1 (system I) and 1:9 (system II) molar ratios. The presence of oxygen from the ether group in the main chain of ethyl acetate leaves the electron cloud more distributed, resulting in poor characteristics of RDF in the first solvation shell (Fig. 3a). However, long-range electrostatic interactions prevail over the sharper second solvation shell with a CN of 1.02, whereas the presence of lone pair and the inductive effect on oxygen of carbonyl group of ethyl acetate shows higher attraction towards —OH moiety of tannic acid. As a result, a sharp peak is seen in Fig. 3b, wherein the first solvation shell is observed at 2.95 Å with a CN of 0.71 and 0.44 for 1:1 and 1:9 M mixtures, respectively.

Figs. S8–S10 of Supplementary material reveal the RDF of catechin and epicatechin with respect to the oxygen atoms from ether moiety and carbonyl moiety of ethyl acetate. The interaction of various hydrogen from the —OH group of catechin and epicatechin with two types of oxygen in ethyl acetate is reflected in RDFs. Figs. S8 and S9 depict the low-intensity RDF implying the submissive character for H-bond formation ether oxygen of ethyl acetate, whereas Figs. S10 and S11 depict the possible H-bond formation involving the —OH moiety of catechin

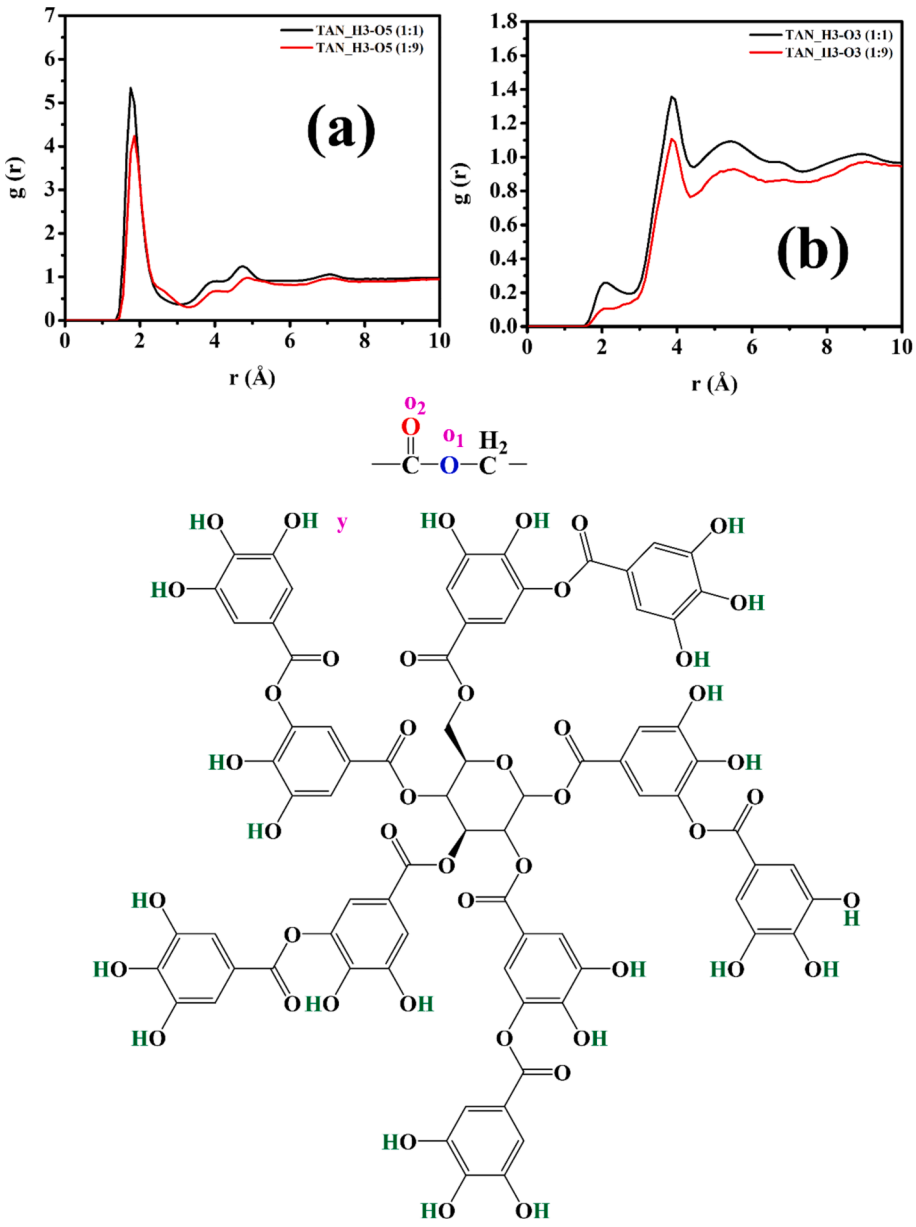


Fig. 3. RDF Tannic acid of ethyl-acetate based system; (a) H3—O5 (denoted by y…o_2) and (b) H3—O3 (denoted by y…o_1).

and epicatechin. This is in line with the revelation of tannic acid solvated in ethyl acetate. However, the solvation shell of trace compounds does not always follow the trail of major compounds, i.e. envelops of RDF increase with a molar ratio of solvent. The other compounds that are found in very small quantities in Guarana seed are procyanidin, theobromine, and theophylline. Despite not having significant extraction possibility, the dominance of carbonyl moiety for the H-bond formation over ether moiety of ethyl acetate is further evident from the RDFs of procyanidin, theophylline, the obromine as given in Figs. S12–S14 of Supplementary materials. The stronger H-bond formation in procyanidin involving —OH moiety of procyanidin with the carbonyl moiety of ethyl acetate is revealed with a *CN* of 0.77 and 0.1 for molar ratios 1:1 and 1:9 respectively, in first solvation shell. Similar *CN* of 0.54 and 0.72 are observed for theobromine and theophylline from Figs. S13a and S14a of Supplementary material. The replacement of methyl moiety by hydrogen in the N2 position of caffeine generates the structure of theophylline. However, this replacement marks significant changes in the RDF, as can be seen in Fig. S13 of Supplementary material. The interactions are more evident or stronger in Fig. S13a as compared to S11b due to the presence of free oxygen with —CO moiety of ethyl acetate. Hence, more electronegative characteristics for the interaction with the hydrogen atom of —NH moiety as compared to the oxygen of ether moiety. A similar trend is shown by the theobromine wherein the interaction of hydrogen of —NH moiety from the pentagon with two oxygens of ethyl acetate is depicted in Fig. S14a–b of Supplementary material.

3.1.3. Distribution of various components of Guarana seed in DES

Fig. 4 illustrates the RDF among various moieties of caffeine and DES being mixed in 1:1 (system I) and 1:9 (system II) molar ratios, along with the representation of functional moieties in the subplots. The oxygen atoms of caffeine can interact with the hydrogen of —OH moieties more intensely than their nitrogen counterpart, as shown in Fig. 4a. The first solvation shell is detected at 2.75 Å with *CN* of 0.7 and 0.49 for 1:1 and 1:9 M ratios, respectively. Additionally, Fig. 4b–d denotes the distribution of various nitrogen moieties with respect to the hydrogen of the —OH moiety of glycerol. A general observation from the three subplots refers to the increase in nonbonded interaction among the said moieties

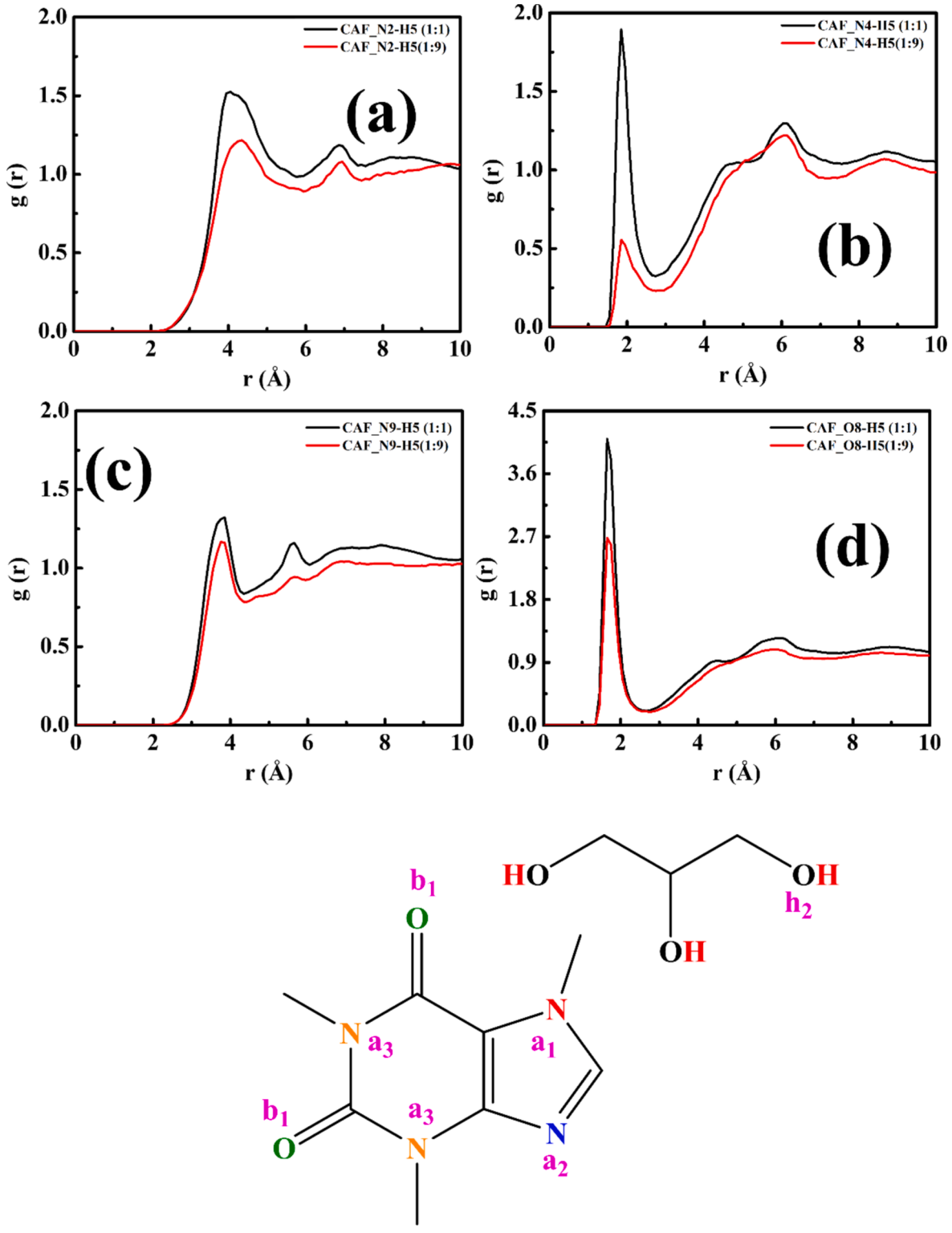


Fig. 4. RDF of caffeine in DES based system: (a) N2—H (denoted by $a_1 \ldots h_2$), (b) N4—H (denoted by $a_2 \ldots h_2$), (c) N9—H (denoted by $a_3 \ldots h_2$) and (d) O9—H (denoted by $b_1 \ldots h_2$).

with the enhanced molar ratio of caffeine. The higher amount of caffeine (i.e. 1:1 mixture) creates a bigger number of available nitrogen and thus produces a bigger distribution envelope. The sharp nature of the RDF peak in the first solvation shell indicates stronger vdW interaction. Hence, the first solvation shell from CAF_ETH_N2 (denoted by a1) resides in 5.85 Å, whereas the same for CAF_ETH_N4 (denoted by a_2) and CAF_ETH_N9 (denoted by a_3) are 2.85 and 4.45 Å respectively. Among the four nitrogens in caffeine, N2 (belonging to CAF_ETH_N2) influences a larger area and thus has CN of 9.33 and 5.7 for 1:1 and 1:9 M mixtures, respectively. However, the sharpness of RDF denotes the intensity of interaction among moieties. In that sense, the presence of immediate —CH_3 with N2 and N9 leaves the respective electron cloud less exposed due to steric hindrance. However, the electron cloud of N4 is relatively free to have non-bonded interaction with hydrogen of —OH of glycerol

in DES. However, for all nitrogen-related RDFs, the second solvation shell is prominent, implying primarily long-range interactions. The peak values of RDF conclude that oxygen and nitrogen (N4) primarily establish nonbonded interaction with DES.

Fig. 5 shows the RDF curve between hydrogen of —OH moiety of tannic acid and oxygen and chlorine anion of ChCl moiety of DES. The nature of RDF involving the oxygen atom of ChCl reveals negligible H-bond formation capability with the —OH moieties of tannic acid (Fig. 5a). However, its interaction with Cl^- anion shows a sharp peak in the RDF given in Fig. 5b. The lone atom-based chloride anion is relatively free to form a H-bond with the —OH moiety of tannic acid. The electron cloud of Cl^- navigates towards any available electropositive element (here, it is the hydrogen of —OH moiety of tannic acid) by making stronger electrostatic interactions with higher intensity RDF seen in Fig. 5b.

The interactions of hydrogen of —OH moieties with catechin and epicatechin are studied with respect to oxygen and chlorine anion of ChCl. The availability of an electron cloud of Cl- leads to the formation of sharp peaks in RDFs, signifying stronger interaction among the atoms. Figs. S15 and S16 in Supplementary material suggest stronger electrostatic interaction with higher RDF intensity. However, the presence of consecutive —OH moieties (Figs. S15b–c and S16a–b of Supplementary material) lead to the formation of intramolecular H-bonding with even more RDF value. For the molar ratio of 1:9, the amount of chloride anion

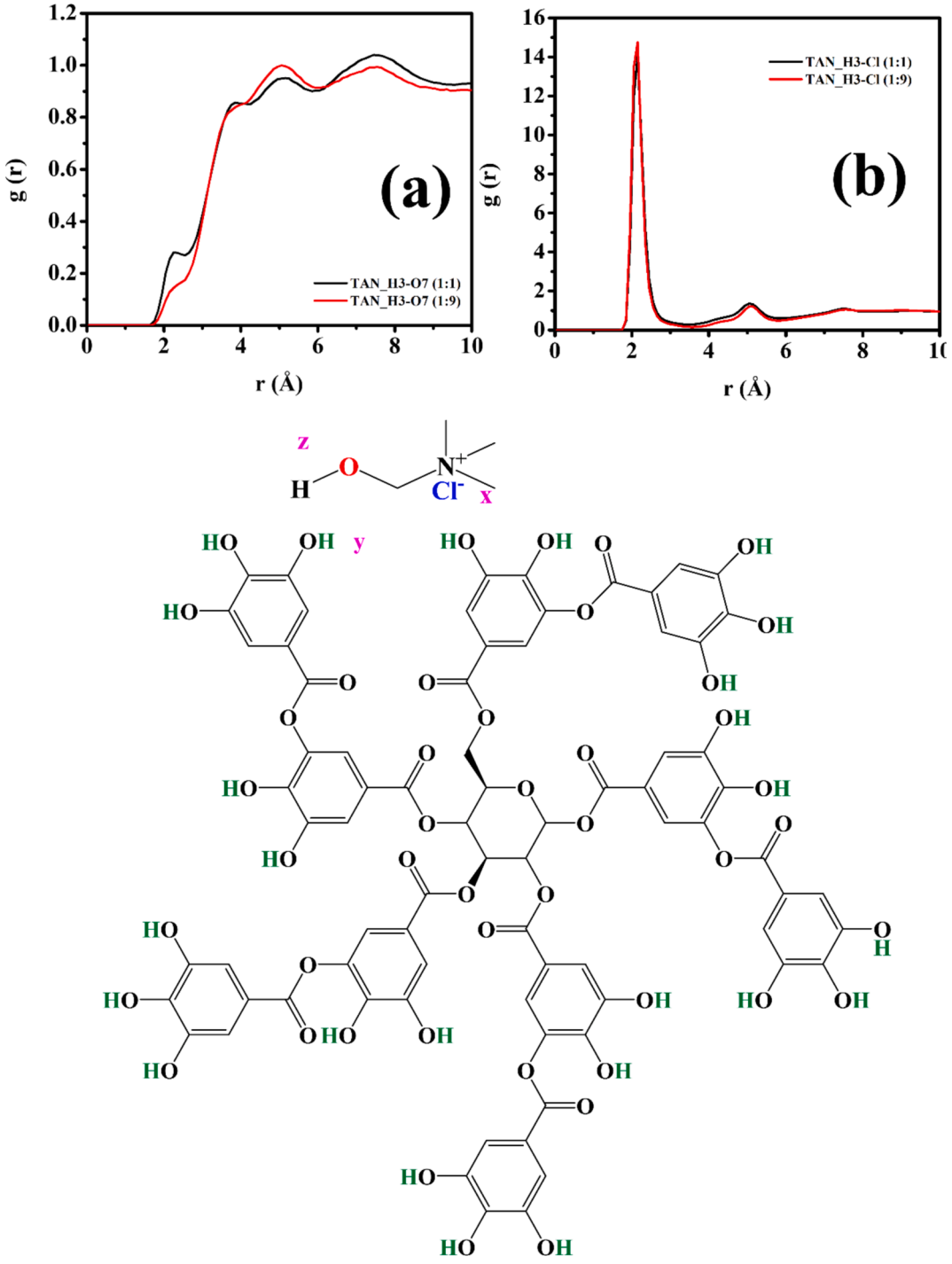


Fig. 5. RDF of Tannic acid in DES based system: (a) H3—O7 (denoted by y…z) and (b) H3—Cl (denoted by y…x).

increases and subsequently available —OH decreases the RDF values (Figs. S15a and S16c) in comparison with the 1:1 M ratio. However, an oxygen atom of ChCl follows a similar trend of interaction as tannic acid (Figs. S17 and S18 of Supplementary material). The other trace elements present in Guarana seeds are procyanidin, theophylline and theobromine. The interaction between hydrogen of —OH moieties of procyanidin and oxygen and Cl^- atoms of ChCl shows similar trends as that of tannic acid i.e. stronger electrostatic interaction between —OH and Cl^- anion due to the presence of denser electron cloud density (Fig. S19 of Supplementary material). The RDFs of theophylline and theobromine with DES are displayed in Figs. S20 and S21, respectively of Supplementary material. Stronger electrostatic interaction with the chloride anion of DES and —NH moiety of theophylline and theobromine, as well as negligible interaction between the oxygen of ChCl and —NH moiety of theophylline and theobromine are observed from Figs. S20 and S21, respectively, of Supplementary material.

Comparing the RDF results in ethanol, ethyl acetate, and DES shows that ethanol has the highest interaction with caffeine and tannic acids at a molar ratio of 1:1. To understand the better interactions, an additional 100 ns simulation is conducted. The RDF results are plotted in Figs. S22

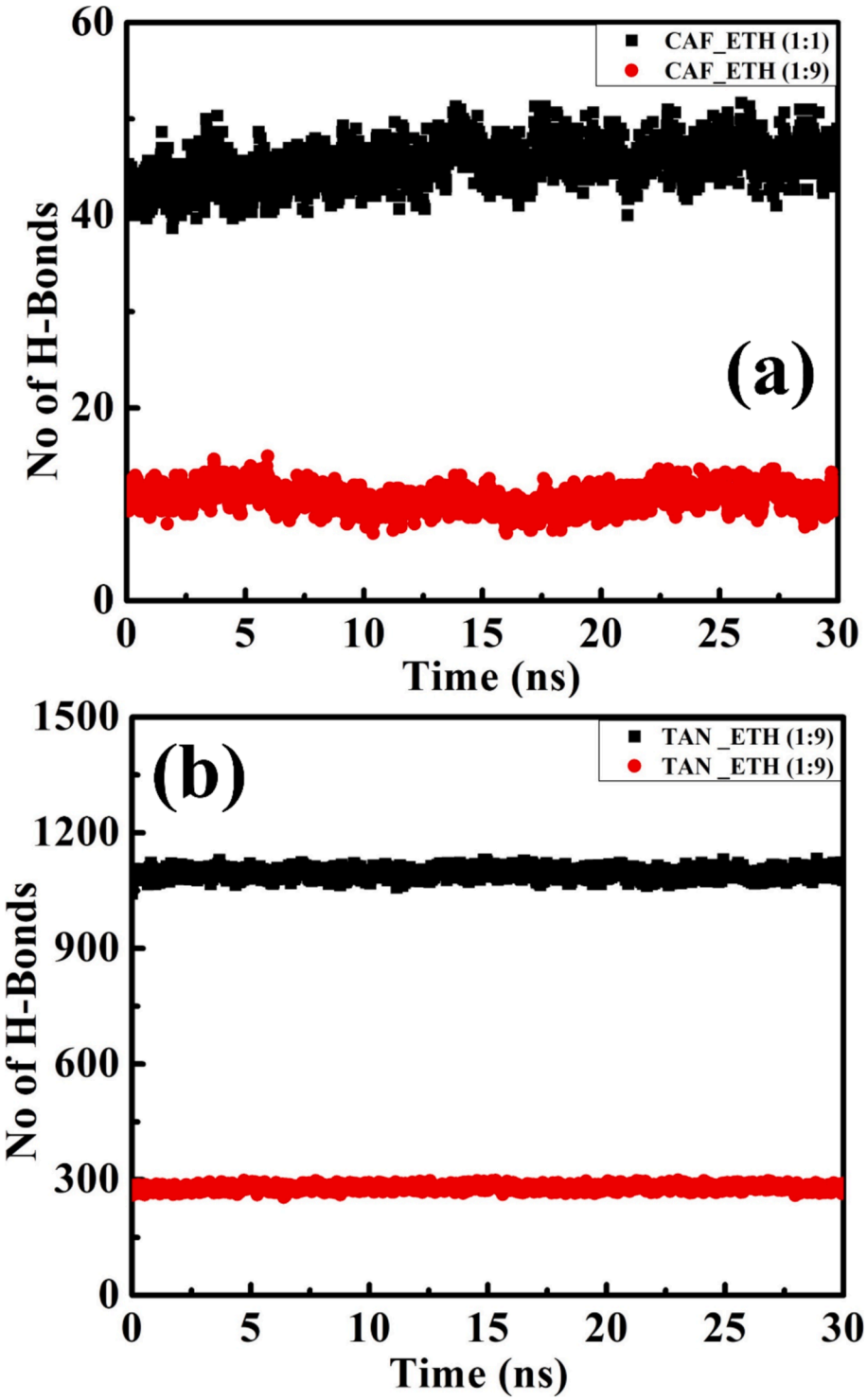


Fig. 6. (a). Number of Hydrogen bonds in caffeine in ethanol-based system and (b) Number of Hydrogen bonds in tannic acid in ethanol-based system.

and S23 of SM. The solvation shell of different moieties of caffeine with ethanol shows similar results to Fig. 1. The solvation shell of different moieties of tannic acids to ethanol shows similar results to Fig. 2. This suggests the changing of simulation time does not affect the RDF's result.

3.2. Evolution of hydrogen bonding network

A comparative plot of the H-bond formation of caffeine and tannic acid in various solvents is given in Fig. 6. H-bond plays a dominant role among various nonbonded interactions and is prominent from the nature of the subplots of Fig. 6. Herein, the numbers of H-bonds significantly increase in the 1:1 mixture compared to the 1:9 mixture because of the increased availability of H-bond sites. The macromolecular structure of tannic acid provides numerous H-bond sites, and hence, a multifold increment in the number of H-bonds than caffeine has been observed in ethanol (Fig. 6a–b). However, marginal fluctuation of H-bond has been observed over time in ethanol solvent, implying the stable H-bond network of caffeine and tannic acid in ethanol. The stability of the H-bond network over time is further visible for minor compounds (catechin, epicatechin, procyanidins, theophylline and theobromine) as depicted in Figs. S24a–b, S25a–b, and S26, respectively of Supplementary material. However, this H-bond network of minor components reveals a lower number of H-bond formation than caffeine and tannic acid. There are two primary reasons for such behaviour. Firstly, the trace amount of presence of these components yields a number of lower H-bonds. Additionally, the affinity of oxygen and nitrogen-containing moieties is higher to form H-bond with ethanol, leading to the selective extraction of caffeine and tannic acid.

The absence of hydrogen atoms associated with the electronegative moieties leaves ethyl acetate without H-bond with caffeine. However, a large number of H-bond networks can be observed between tannic acid and ethyl acetate, as revealed in Fig. 7. Since H-bond plays a crucial role in extraction, this explains the lower extraction performance of ethyl acetate in the extraction of caffeine and tannic acid from Guarana seed. The H-bond network for minor compounds (catechin, epicatechin and procyanidin) shows a similar trend as shown in ethanol and DES with lesser values indicating lower formation number of H-bond with ethyl acetate as seen in Figs. S27a–b and S28 of Supplementary material respectively.

The stability of the H-bond network for tannic acid and caffeine further corroborates for DES. As revealed from the RDFs, chloride ions from ChCl and oxygen moieties from glycerol can make an H-bond network; such evolution over time has been visualized in Fig. 8 and 9. The H-bond network in DES follows a similar trend with respect to the average number and molar ratio of solvent to that of ethanol. The H-bond network of minor components suggests a lower number of H-bond formation than caffeine and tannic acid. For minor compounds (catechin, epicatechin, procyanidins, theophylline and theobromine), the H-bond network with ChCl shows a scattered trend showing almost negligible H-Bond. However, their interaction with glycerol shows a comparatively better H-bond network but with lower values as compared to caffeine and tannic acid, as seen in Figs. S29a–b, S30a–b, S31a–b, S32a–b and S33a–b of Supplementary material respectively. Fig. S34 of Supplementary material illustrates the number of H-bond formations between the HBD and HBA of DES. Since the H-bonding between HBD and HBA is the key interaction of DES formation, such stability over a prolonged period reveals the structural integrity of DES. The effect on the molar ratio of DES and Guarana seed extract has the reverse effect on the evolution of the H-bond network of DES constituents. With the increase in solute (i.e. Guarana seed extract), the functional moieties of HBA and HBD rush to form H-bonding with the solute components (caffeine, tannic acid, theobromine, theophylline, catechin, epicatechin and procyanidins). This shows a stronger affinity of solute towards solvent than the intramolecular H-bond affinity of DES. This

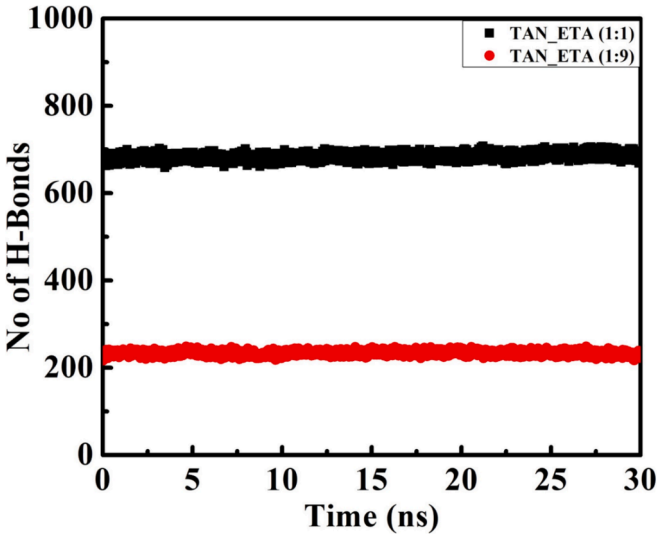


Fig. 7. Number of Hydrogen bonds in tannic acid in ethyl acetate based system.

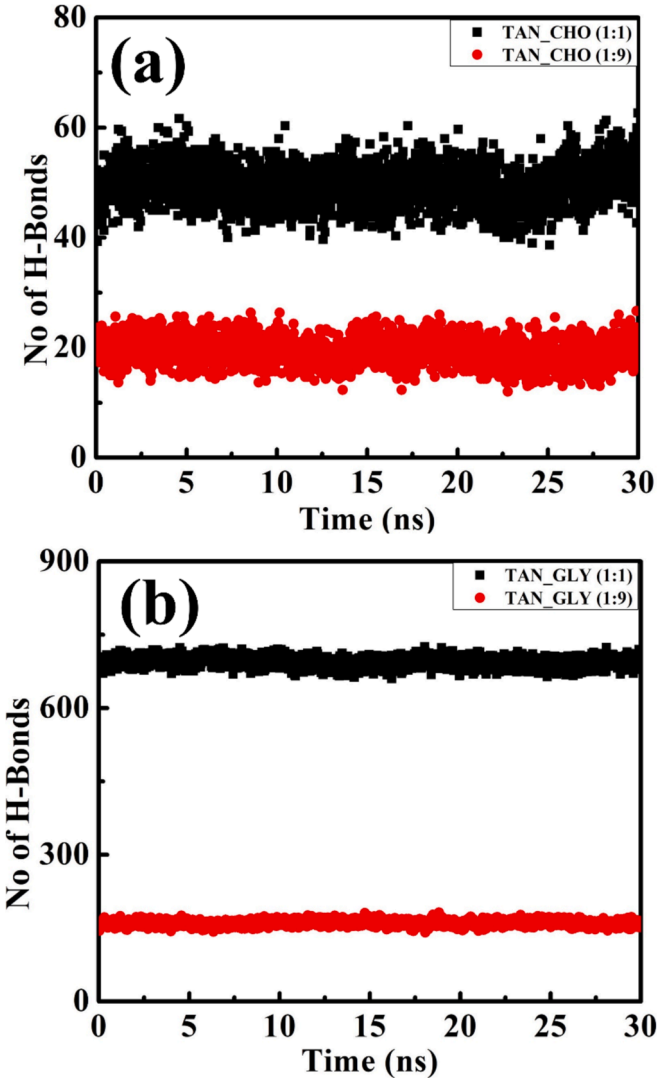


Fig. 8. (a). Number of Hydrogen bonds in Tannic acid in Choline chloride-based system and (b) Number of Hydrogen bonds in tannic acid in glycerol based system.

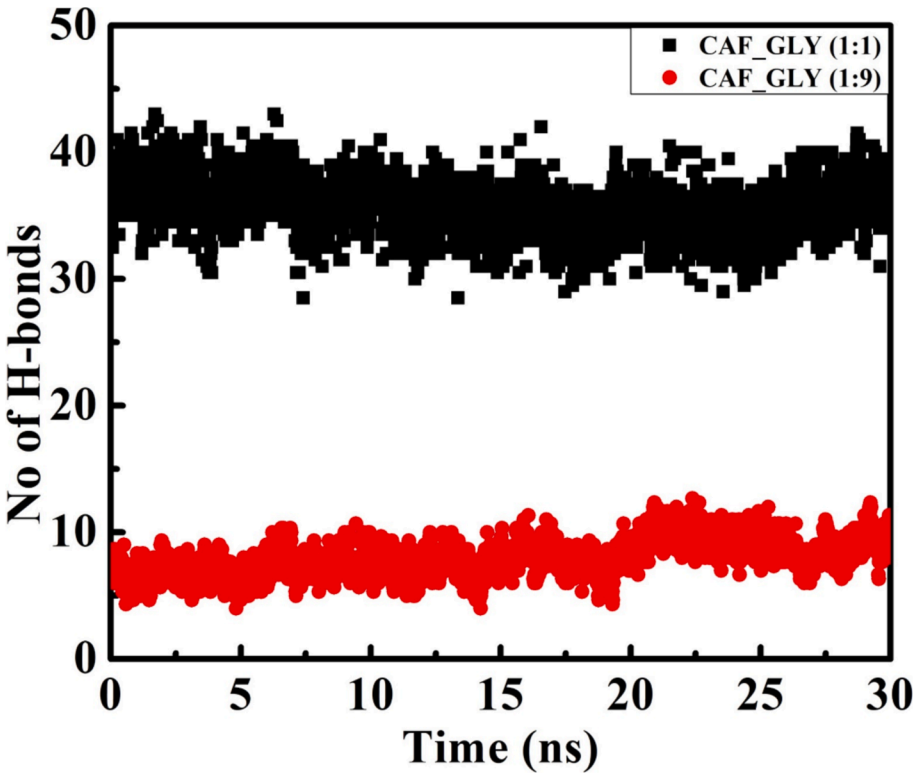


Fig. 9. Number of Hydrogen bonds in Caffeine in glycerol based system.

further facilitates the enhanced extraction using DES and adds one more potential system to an already established state-of-the-art extraction agent [44]. However, the structural integrity of DES remains intact from the constant H-bond network over a prolonged time period.

Comparing the three solvents, the ethanol system (1:1) is forming more H-bonds with caffeine compares to the other system. These findings are also validated by the RDF results. The experimental literature suggests that the yield of caffeine using ethanol is 18.4 %, and the yield of caffeine in DES is 4.93 % [25,45]. These findings are also validated by the simulation results. To enhance comprehension of H-bonding dynamics, a 100 ns simulation run is carried out for the ethanol (1:1) system. Fig. S35 shows the average number of H-bonds with caffeine to ethanol and tannic acids to ethanol for a longer period. However, the average number of H-bonds between the caffeine to ethanol and tannic acids to the ethanol is the same as the 30 ns result. This suggests the system is in equilibrium. So, changing in time doesn't affect the H-bonding dynamics. During the H-bonding calculations, the strength of the H-bonding is calculated in quantum mechanics/ mixed mechanics (QM/MM) MD. The H-bond strength is not calculated in this study due to the limitations of applying the QM/MM approach to a system of this scale. The complexity and size of the molecular system under investigation pose significant challenges to the effective implementation of QM/MM calculations. While quantifying selected hydrogen bonds could provide valuable insights, the large size of certain structures in this study makes this approach impractical with the available computational resources.

3.3. Means square displacement (MSD)

3.3.1. Mobility of caffeine and tannic acid in solvents

The distributions of leading functional moieties for the extraction of caffeine and tannic acid are established in the RDF. However, their time-resolved mobilities are visualized from the mean squared displacement (MSD) of functional moieties. The diffusive behaviour of major components of Guarana seeds, i.e. caffeine and tannic acid, are estimated from the MSD plots. Hence Figs. 10 and 11 illuminate the MSD of functional moieties in ethanol, ethyl acetate and DES over 1:1 and 1:9 M ratios. A general observation from Fig. 10 reveals about two diffusive behaviours of electronegative atoms of caffeine in ethanol. Like the RDF and H-bonding, the molar ratio of solvent plays a crucial role in the mobility of molecules, leading to selective extraction. Fig. 10 suggests that the MSD of caffeine in a 1:1 M ratio reduces approximately 100-fold to a 1:9 M ratio. The large amount of solute in a 1:1 M ratio produces multifold increments in the H-bonding, leading to the restricted movement of components. However, for both cases, two regions of the MSD profile, i.e. monotonous increment and concave downward, are observed. For the 1:1 M ratio case, the changeover happens in the range of 27 ns. The lower number of H-bonding in the 1:9 M ratio systems reduce the changeover to approximately 24 ns. During the monotonous increment regime, caffeine produces a subdiffusive behaviour yielding a β value less than unity. The subdiffusive behaviour suggests the random motion of the functional moieties in the ethanol.

Fig. 11 reveals the diffusive behaviour of caffeine in DES estimated from the MSD curve. It generally shows two different diffusive behaviour for all electronegative functional moieties. The linear curve is observed for nitrogen and oxygen moieties for a molar ratio of 1:1, whereas two subsections (monotonously increasing followed by concave downward) are observed for a molar ratio of 1:9. The diffusive nature closely resembles the compound's mobility in ethanol. Slow diffusion of oxygen moieties of caffeine is observed till 23 ns for a 1:1 M ratio, followed by the significant superdiffusive behaviour of oxygen moieties (Fig. 11g). The superdiffusive behaviour is also observed for a 1:9 M ratio (Fig. 11h). Fig. 11 suggests that the MSD of caffeine in a 1:1 M ratio reduces approximately 100-fold to a 1:9 M ratio. The large amount of solute in a 1:1 M ratio produces multifold increments in the H-bonding, leading to the restricted movement of components.

Fig. 12 reveals the MSD curves of different moieties of the tannic acid in the ethanol system (both 1:1 and 1:9 M ratios). The subdiffusive behaviour has been observed for the 1:1 M ratio of Guarana seed extract and ethanol. The relatively dense packing, as well as the

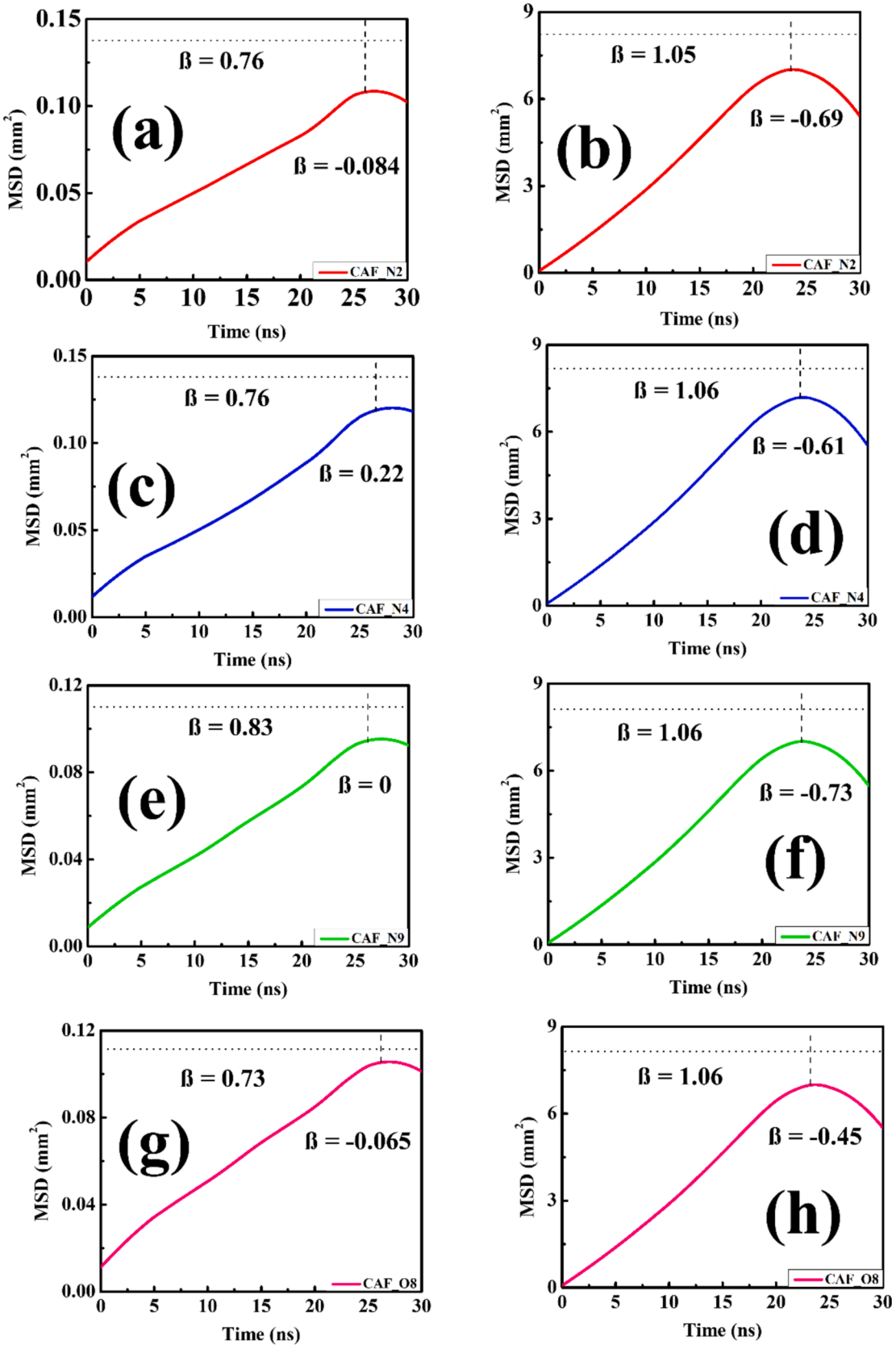


Fig. 10. MSD curve of caffeine in ethanol-based system I (1:1): (a) N2, (c) N4, (e) N9, (g) O8, system II (1:9): (b) N2, (d) N4, (f) N9, (h) O8,

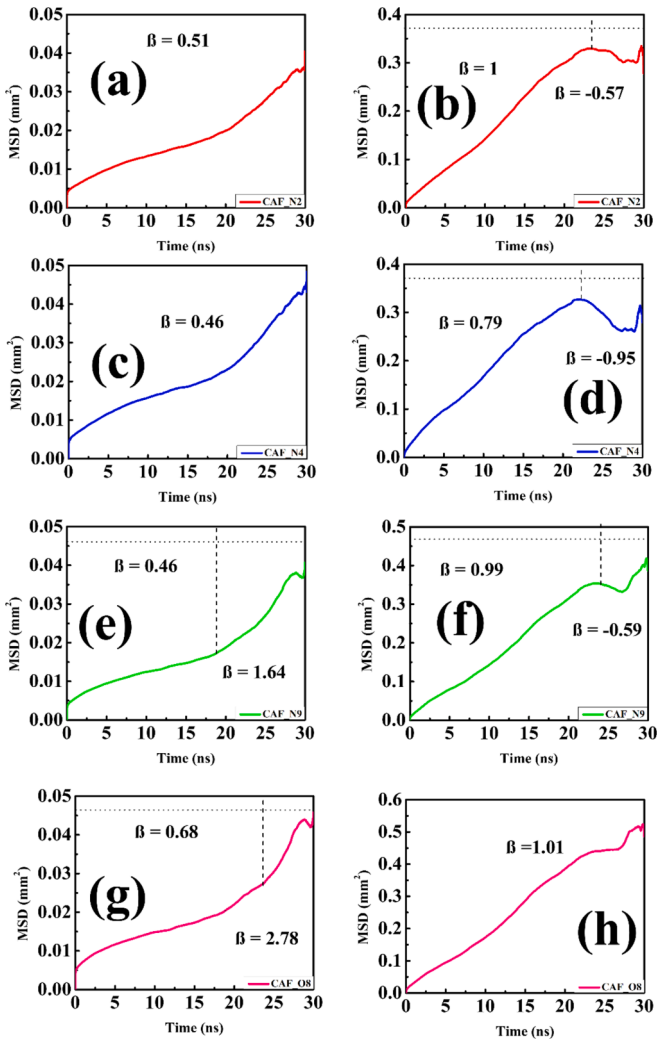


Fig. 11. MSD curve of caffeine in DES-based system I (1:1): (a) N2 (c) N4 (e) N9 (g) O8, system II (1:9): (b) N2, (d) N4, (f) N9, (h) O8,

macromolecular geometry of tannic acid, hinder the movement through solvent, leading to the subdiffusive behaviour. However, an increased amount of solvent reduces the density of H-bonding, and tannic acid shows superdiffusive behaviour in a 1:9 M ratio mixture. The super-diffusive behaviour accelerates the extraction, and tannic acid can be extracted faster over caffeine. Fig. 13 reveals the MSD plot of tannic acid in ethyl acetate, which is primarily subdiffusive in nature. However, the random motion of —OH moiety increases with the increment of solvent ratio leading to a β value of 0.96 (Fig. 13b). The MSD profiles of tannic acid in DES produce slow diffusion leading to a β values of 0.65 and 0.63 for 1:1 and 1:9 M mixtures (Fig. 14). The larger number of H-bonding creates significant hindrance in the movement and increment of solvent content increases the slowness in movement.

Fig. S36 shows the MSD of the N2 moiety of caffeine and H3 moiety of tannic acids in the ethanol (1:1) system for 100 ns simulation. The MSD curve of caffeine and tannic acids shows the subdiffusive behaviour throughout the dynamics. However, the slope of dynamics is higher compared to the 30 ns simulations. This is because longer simulations allow for better statistical averaging, which reduces errors.

3.3.2. *Mobility of minor components in solvents*

Figs. S37 and S38 illustrate the MSD curve of different oxygen moieties of catechin in the ethanol system (for both 1:1 and 1:9 M ratios, respectively). Two subsections are significantly observed for a 1:1 M ratio, showing a subdiffusive nature. However, opposite behaviour is observed for a 1:9 M ratio wherein two subsections, first till 22 ns, show subdiffusive behaviour for a further 8 ns. The nature shows sharp elevation showing super diffusive behaviour of moieties with β values greater than 2. The MSD profile for catechin in ethyl acetate for 1:1 and 1:9 M ratio is reflected in Figs. S39 and S40 of Supplementary material, respectively. Three subsections are revealed for the system of molar ratio 1:1, mainly signifying slow diffusion throughout timeframes. However, the linear nature of the graph for the 1:9 M ratio reveals the subdiffusive behaviour of different hydrogen moieties in ethyl acetate. Figs. S41 and S42 reveal the MSD plot of catechin in DES. MSD curve for 1:1 M ratio here shows two trends, linear and concave downward, decreasing with less β value. However, completely opposite can be seen for molar ratio 1:9 where two subsections, first subdiffusive followed by superdiffusive with β values greater than 1.

Figs. S43 and S44 reveals the MSD curve of different moieties of epicatechin in the ethanol system (for both 1:1 and 1:9 M ratios respectively). MSD curve for 1:1 M ratio shows similar trends for all moieties where they show subdiffusive nature before the changeover followed by the superdifsfusive nature. Thereafter, different subsections can be seen for 1:9 M ratio with β value less than 1, basically signifying slow diffusion all over. The MSD plots for epicatechin in ethyl acetate for both the molar ratios are reflected in Fig. S45 and S46. For molar ratio 1:1 two trends i.e. linear and concave downward is observed with β value less than 1 signifying slow diffusion whereas 1:9 M ratio shows superdiffusivity behaviour. Figs. S47 and S43 show MSD plots of different hydrogen moieties of epicatechin in DES system for 1:1 and 1:9 M ratios respectively. Fig. S47 shows slow diffusion whereas S48 shows superdiffusive nature.

Figs. S49 and S50 reveals the MSD curve of different moieties of procyanidin in the ethanol system (for both 1:1 and 1:9 M ratios respectively). The subdiffusive behaviour has been observed for 1:1 M ratio of Guarana seed extract and ethanol for Fig. S49a–b. The dense packing of procyanidin hinders the movement through solvent leading to the subdiffusive behaviour whereas for Fig. S49c superdiffusive behaviour is observed. Thereafter, increased amount of solvent for 1:9 M ratio reduces the density of H-bonding resulting in superdiffusive behaviour. Fig. S51 reveals the MSD plot of procyanidins in ethyl acetate which is primarily subdiffusive in nature. Furthermore, the random motion of —OH moiety increases with increment in solvent ration leading to β value of 0.86 (Fig. S51b). The MSD profile of procyanidin in DES produces slow diffusion leading to a β values of 0.61 for both the molar mixtures (Fig. S52).

Figs. S53 and S54 reveal the MSD curves of different moieties of theophylline in the ethanol system (for both 1:1 and 1:9 M ratios respectively). The oxygen moiety of theophylline shows subdiffusive behaviour throughout time whereas nitrogen moieties show two sub-sections i.e. subdiffusive till 16 ns followed by superdiffusive behaviour till 30 ns for 1:1 M ratio. However, for molar ratio 1:9 the behaviour of all the moieties in first subsection till 23 ns shows superdiffusive behaviour and subdiffusive behaviour for next 7 ns. Fig. S55 reveals the MSD plot of theophylline in ethyl acetate which is primarily subdiffusive in nature. At the same time, for 1:9 M ratio as the amount of solvent increases the nature of MSD is superdiffusive in nature with value of 1.01 (Fig. S55b). The MSD profile of theophylline in DES produce slow diffusion for 1:1 M ratio. Moreover, for 1:9 M ratio two subsections are observed first showing subdiffusive behaviour followed by the super-diffusive behaviour for second subsection with β values of 0.39 and 1.84 for 1:1 and 1:9 M mixtures respectively (Fig. S56b).

Figs. S57 and S58 reveal the MSD curve of different moieties of theobromine in the ethanol system (for both 1:1 and 1:9 M ratios, respectively). The different moieties of theobromine show subdiffusive behaviour throughout time for 1:1 M ratio. However, for molar ratio 1:9 the behaviour of all the moieties shows superdiffusive behaviour with logarithmic slope greater than unity. Fig. S59 reveals the MSD plot of theobromine in ethyl acetate which is primarily subdiffusive in nature for both molar ratios with two subsections for each. The MSD profiles of

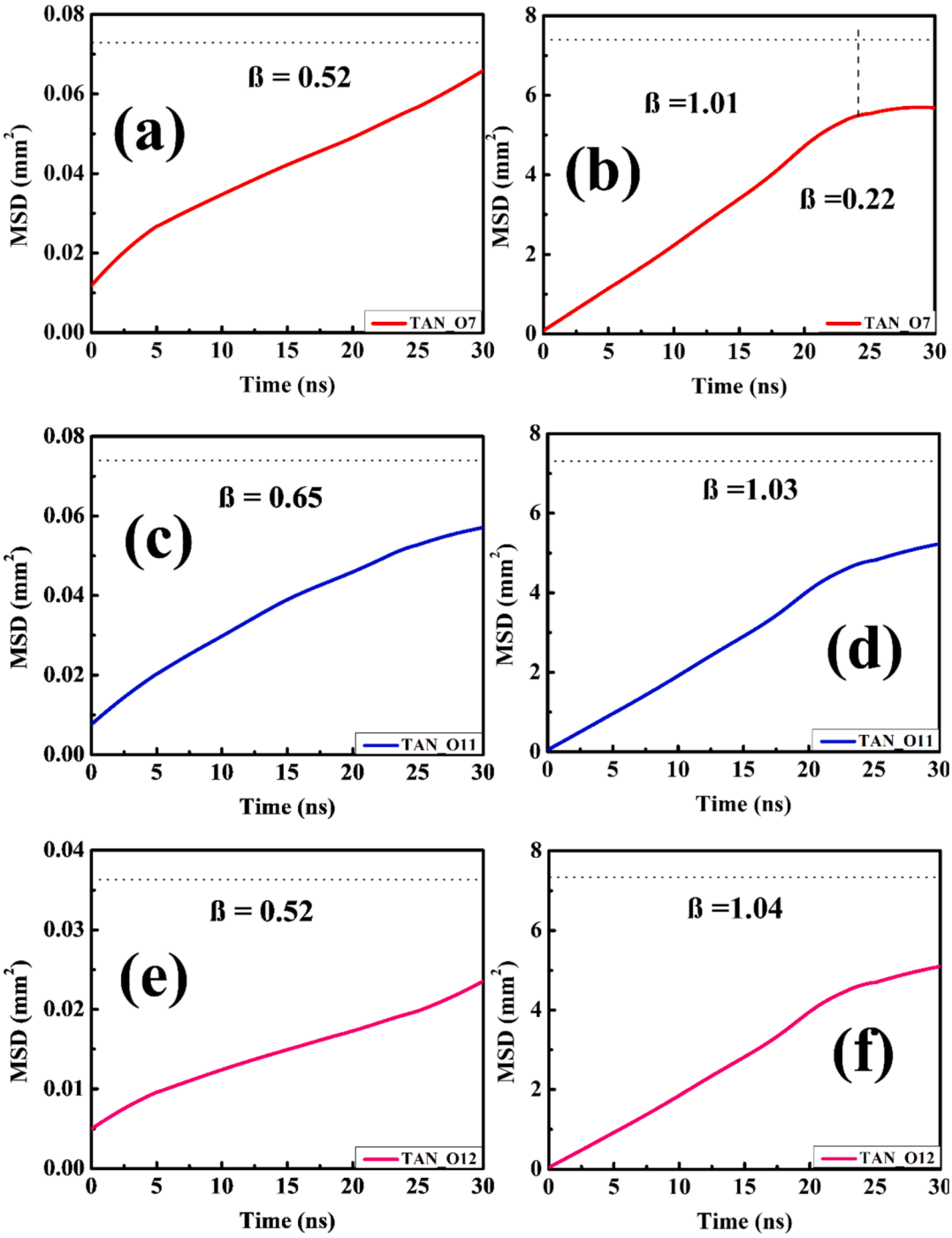


Fig. 12. MSD curve of Tannic acid in ethanol-based system I (1:1) (a) O7, (c) O11 and (e) O12, system II (1:9): (b) O7, (d) O11 and (f) O12.

theophylline in DES produce slow diffusion till 23 ns for a 1:1 M ratio, and for a further 7 ns, the behaviour changes to a superdiffusive character. However, for a 1:9 M ratio, subdiffusive behaviour is observed throughout time with β values of 0.71 (Fig. S60b).

4. Conclusion

The extraction of major and minor components of Guarana extract and its interaction with different solvents, i.e. ethanol, ethyl acetate and ChCl/glycerol-based DES, are investigated and analysed in this study with the help of all-atoms MD simulation. Useful insights into the arrangement of the system varying the molar composition of solute and solvent and its influence on the properties of components of Guarana extract and solvents are obtained and thus broader understanding of extraction is elucidated. The analysis of hydrogen bonding reveals the structural integrity of DES over the course of the simulation. The H-bonding results show that the 1:1 M ratio forms more H-bonds compared to 1:9. The greater number of possible H-bonding sites facilitates the close association of tannic acid and caffeine with ethanol and DES over the minor components. This further justifies the higher amount of extraction efficiency of caffeine and tannic acid using ethanol and DES. A large envelope of RDF in the first solvation shell has been observed in the ethanol solvated medium, implying the dominance of short-range interactions such as H-bonding among ethanol and solute moieties. However, an equimolar mixture of solute and ethanol increases the association strength and hence raises the extraction capability. Stronger

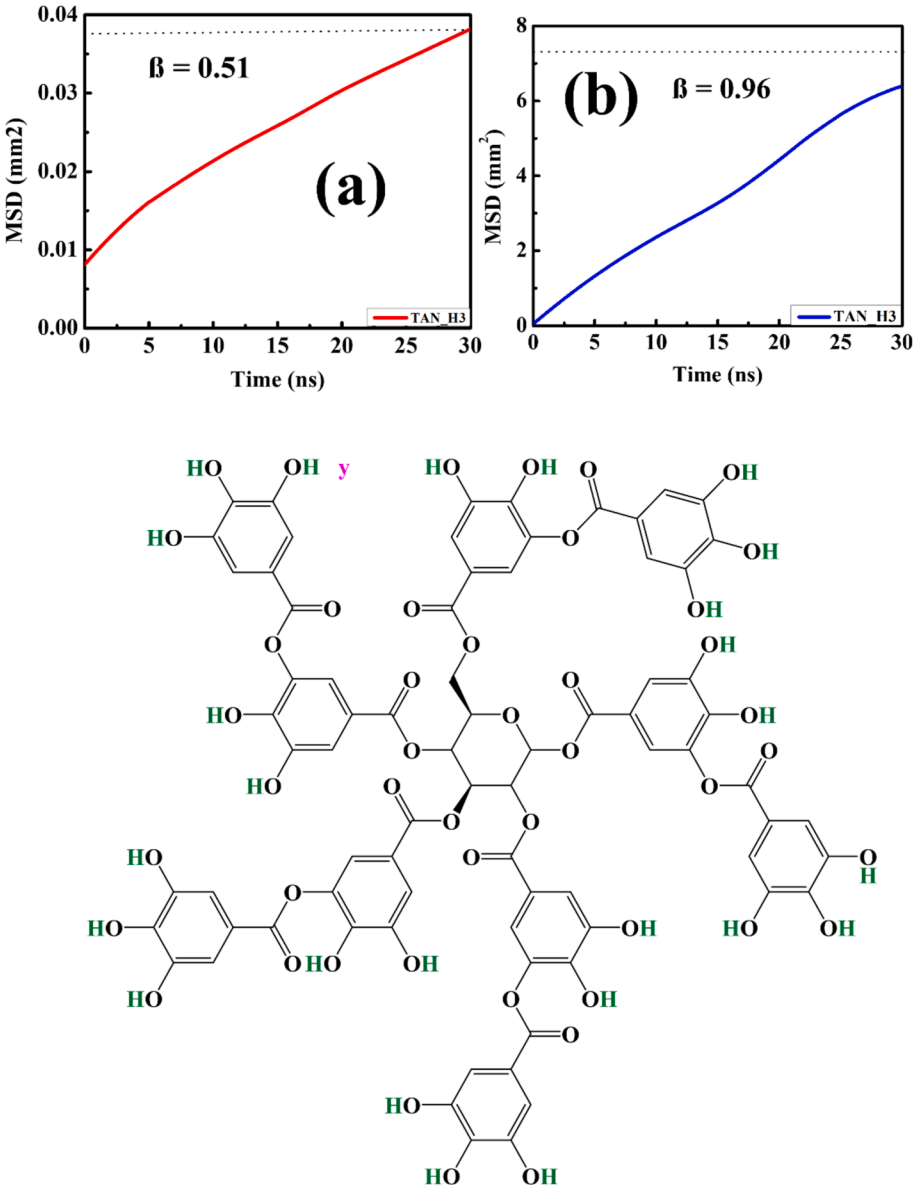


Fig. 13. MSD curve of Tannic acid in ethyl acetate-based system: (a) H3 (system I) (denoted by y), (b) H3 (system II) (denoted by y).

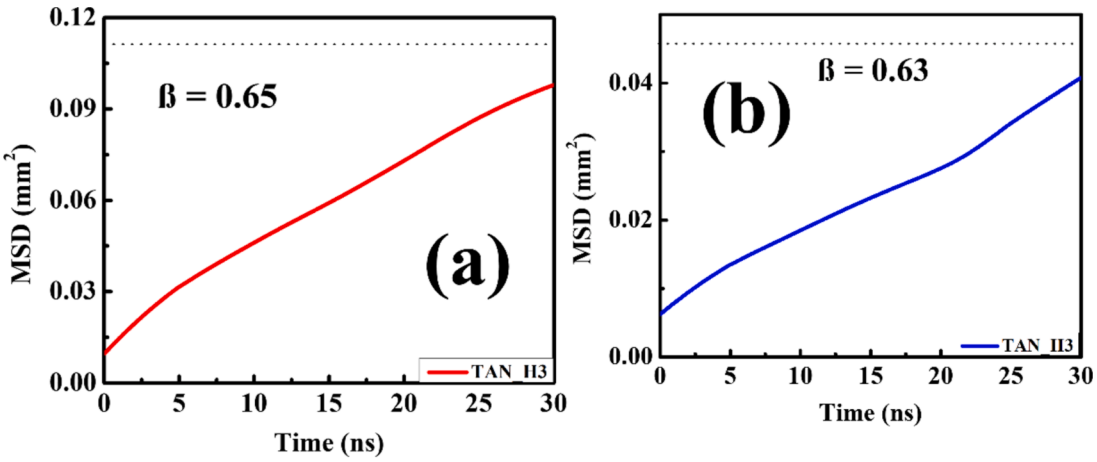
interaction is being observed among oxygen moieties of caffeine and tannic acid with the —OH moiety of ethanol. The RDF analysis further indicates ethyl acetate as the weakest solvent among the considered three solvents for caffeine and tannic acid extraction. The mobility of components is elucidated by the analysis of MSD. Strong nonbonded interaction hinders the mobility of the components, as in the case of ethanol. This raises the extraction probability of ethanol and justifies the wider usage of caffeine extraction. The Guarana seed components in ethyl acetate solvent mostly adheres superdiffusive nature leading to the difficulty in encapsulation or nonbonded interaction. Hence, the poor extraction capability of ethyl acetate has been justified. The ethanol (1:1) system shows stronger interactions compared to the other solvents. To better understand the dynamics, the simulation of the ethanol (1:1) system ran for up to 100 ns. The key results suggest that the structural properties do not change over time. However, the transport property, such as MSD, shows subdiffusive behaviour but the slope of the MSD curve increases, which is due to better statistical averaging. Future work could explore the application of these findings to optimize extraction processes for other plant-based compounds. This study contributes to the scientific community by providing a molecular-level understanding of solvent–solute interactions, which can guide the development of more efficient and selective extraction methods for natural products.

CRediT authorship contribution statement

Abhinav Tidke: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Rontu Das:** Visualization, Validation, Formal analysis. **Uday S. Annapure:** Project administration. **Debashis Kundu:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



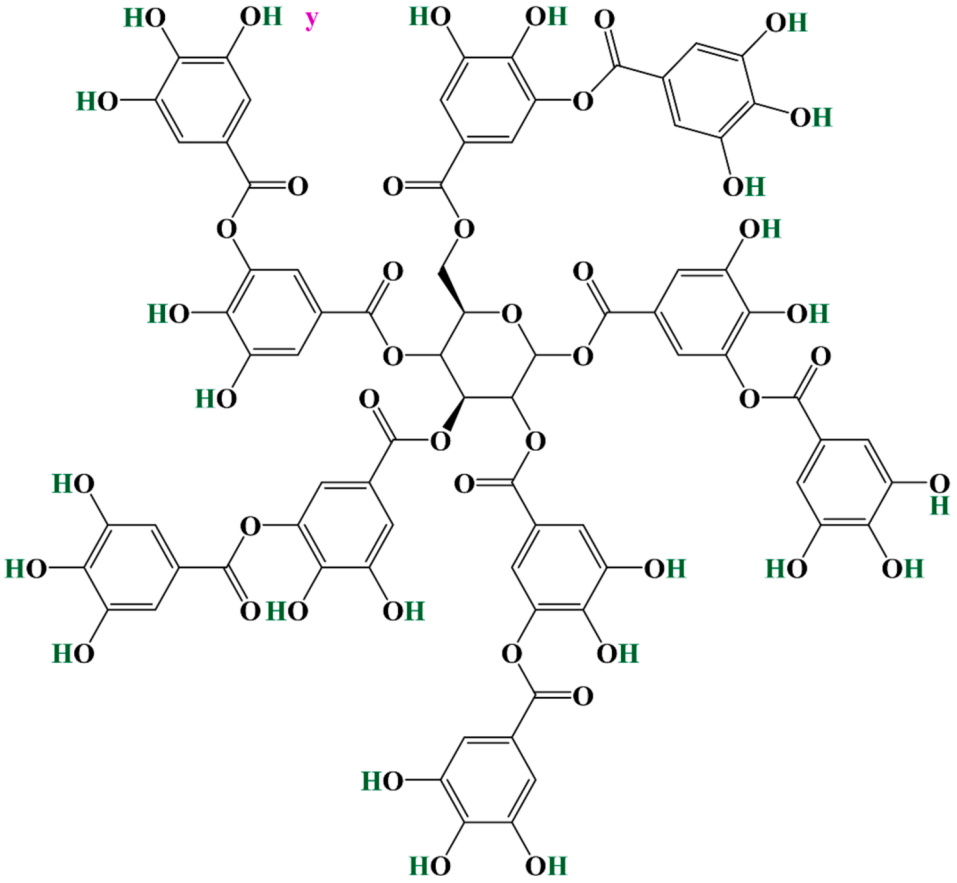


Fig. 14. MSD curve of Tannic acid in DES based system: (a) H3 (system I) (denoted by y), (b) H3 (system II) (denoted by y).

Appendix A. Supplementary material

Detailed information about initial system, RDF plot, MSD plot and Hydrogen bonding plots of minor components of Guarana seed extract are given. Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2024.126462.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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